KENYATTA UNIVERSITY

EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE IN

MECHANICAL ENGINEERING

EMM 214: ELECTRICAL ENGINEERING II

2ND SEMESTER 2020/2021 TIME: 2 HOURS

INSTRUCTIONS:

(i) This paper consists of FIVE questions

(ii) Question ONE is compulsory and contains 30 marks

(iii) Attempt any TWO of the remaining four questions. Each question is 20 marks

Question One (30 marks)

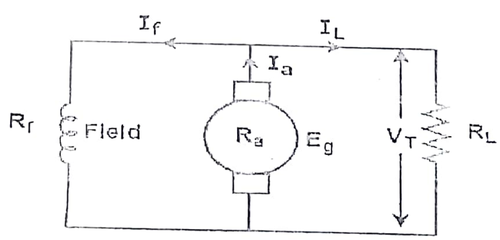
i) What is the principle of operation of a DC Motor? (4 Marks)

ii) A 250V dc motor running at 10 rev/s has an armature resistance of 0.5 Ω. The armature current is 40 A. Determine the torque developed. (4 Marks)

$E = \frac{PZ\phi N}{60A}$ $T\omega = E_b I$

iii) Differentiate between separately excited and self-excited dc generators (4 Marks)

iv) From the circuit diagram of a dc generator derive the voltage equation. (6 Marks)



v) A three-phase induction motor is wound for 4 poles and is supplied from a 50Hz system. Calculate:

a) the synchronous speed

b) the speed of the rotor when the slip is 4%

$\frac{120f}{P} = \frac{120f}{P}$

$I_a R_a || R_f + I_L R_L$ 120 $S = \frac{120f}{P}$

c) the rotor frequency when the speed of the rotor is 600 r/min (6 Marks)

vi) Determine the no-load parameters of a three-phase oil transformer. The main data of the transformer are: P=6000KVA, V1/V2 = 40000/7000V, I1/I2 = 100/500A, Po = 20KW, Io = 5%, winding connection Y/Δ - 11 (6 Marks)

Question Two (20 marks)

i) What is the principle of operation of a DC Generator? (4 Marks)

ii) Explain the three tests performed on transformers. (6 Marks)

iii) A 6 pole, lap wound d.c. motor has 540 conductors with armature resistance of 0.8Ω. Its speed is found to be 1500 r.p.m. when it is made to run light. The flux per pole is 30 mWb. It is connected to a power supply of 250 V. Find the:

a) Induced e.m.f. b) Armature current c) mechanical power d) Armature torque (4 Marks)

iv) At 200V a short-shunt compound generator supplies a current of 120 A. If the armature resistance, Ra = 0.08 Ω, field resistance, Rf = 40 Ω, and the series resistance, Rse = 0.04 Ω. Determine the e.m.f. generated. (6 Marks)

$E = \frac{PZ\phi N}{60}$

$250 - I_a R_a$

Question Three (20 marks)

i) What is the principle of operation of an Induction Motor? (4 Marks)

ii) A 1000KVA single transformer has 0.02 p u resistance, 0.10 p.u reactance and open circuit secondary voltage of 810V. it is connected in parallel with a 250KVA single phase transformer having 0.03 p.u resistance, 0.08p.u reactance and an open circuit secondary voltage of 415V. Find:

$V = IR$

(a) Circulating current at no load
(b) Currents delivered by the secondary of each transformer to a load of 750KVA at 0.8 lagging pf.
(c) KVA and p.f of each transformer (6 Marks)

iii) A 250V, two pole, 60Hz Y-connected wound rotor induction motor is rated at 20 hp. Its equivalent circuit components are:

$R_1 = 0.3\Omega$	$R_2 = 0.2\Omega$	$X_M = 15\ \Omega$
$X_1 = 0.52\ \Omega$	$X_2 = 0.52\ \Omega$	
$P_{mech} = 250W$	$P_{misc} \approx 0$	$P_{mech} = 150W$

Slip = 5%

Calculate:

(a) The line current I_L
(b) The stator copper losses P_{SCL}
(c) The air-gap power P_{AG}
(d) The power converted from electrical to mechanical P_{conv}
(e) The induced torque τ_{ind}
(f) The load torque τ_{load}

$P = I^2R$

$r_0 = \frac{P}{I_0}$

$Z = V/I$

INVOLVEMENT IN ANY EXAMINATION IRREGULARITY SHALL LEAD TO DISCONTINUATION

$P =$ $P = VI$ $V = IR$ $R = \frac{V}{I}$ Po

(g) The overall machine efficiency
(h) The motor speed in revolutions per minute and radians per second (10 Marks)

Question Four (20 marks)

i) Differentiate between the three types of connections of a three phase transformer. (6 Marks)

ii) A 200V d.c. motor develops a shaft torque of 15 Nm at 1200 rev/min. If the efficiency is 80%. Determine the current supplied to the motor. (4 Marks)

iii) A 520V, 50hp, three phase induction motor is drawing 60A at 0.95 pf lagging. The stator copper losses are 2.5kW and the rotor copper losses are 800W. The friction loss is 700W and the copper losses are 1800W. Find:

(a) The airgap power PAG
(b) The converted power Pconv
(c) The output power POUT
(c) The efficiency of the motor

$E_b = T\omega = E_b I_a$ $\quad E_b =$

$E_b =$

(4 Marks)

iv) A separately-excited generator develops a no-load e.m.f. of 150V at an armature speed of 20 rev/s and a flux per pole of 0.10Wb.
Determine the generated e.m.f. when:

$150 = P \cdot Z \cdot 0.1 \times 10^{-3} \times 24$

(a) the speed increases to 25 rev/s and the pole flux remains unchanged
(b) the speed remains at 20 rev/s and the pole flux is decreased to 0.08Wb.
(c) the speed increases to 24 rev/s and the pole flux is decreased to 0.07Wb. (6 Marks)

Question Five (20 marks)

i) Differentiate between the three methods of controlling dc motor speed: Field control method, Armature control method and Ward Leonard. (6 Marks)

ii) Draw the phasor diagram of a transformer with inductive and capacitive load. (6 Marks)

iii) A 100kW, 880V, 100Hz, six pole induction motor has a slip of 6% when operating at full-load conditions. At full-load conditions. The friction and windage losses are 300W and the core losses are 600W. Find the following values for full-load conditions:

(a) The shaft speed n_m
(b) The output
(c) The load torque τ_{load}
(d) The induced torque τ_{ind}
(e) The rotor frequency

$E_g = \frac{PZ\phi N}{60A}$ (8 Marks)

$\omega = 2\pi N$